Question 5.7 of arXiv:2507.00186: all non-bounded-remainder intervals give weak mixing

Literature-implied solution packet

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Status. `literature_implied_answer`; complete, in a stronger all-irrational-rotation form. The one-sided discrepancy theorem needed below is classical (Halász, 1976). The application to the source paper's operator question is the identification recorded in this packet; no claim of a new theorem is made.

1 Source question

The source is Valentin Gillet, *Linear dynamics of random products of operators*, arXiv:2507.00186v3, Question 5.7, printed/PDF page 45. For an irrational rotation R_α, it takes

$$A_1 = [0, b), \qquad A_2 = [b, 1),$$

and asks whether Corollaries 3.22 and 3.23 remain true when $b \notin \mathbb{Z}\alpha$ modulo one. Those corollaries assert topological weak mixing, almost surely in the starting point, for random products of adjoints of multiplication operators satisfying either of the two analytic hypotheses listed there.

Corollaries 3.21, 3.22 and 3.23 hold only for certain intervals of $[0,1)$, since their proofs make use of the computation of the Fourier coefficients of the function $f = \mathbb{1}_{A_1} - \frac{m(A_1)}{m(A_2)}\mathbb{1}_{A_2}$ in order to apply CLTs for the doubling map and for the irrational rotations. Thus, the following question is natural.

Question 5.6. — Do Corollaries 3.21, 3.22 and 3.23 still hold if we replace the two intervals by disjoint Borel subsets of $[0,1)$ such that $A_1 \cup A_2 = [0,1)$ and $m(A_k) > 0$ for $k = 1, 2$?

In the same vein, we have seen in Corollary 3.27 that for some irrational rotations, the random sequence $(T_n(.))_{n\geqslant 1}$ won't be universal. Thus, the following question is also natural.

Question 5.7. — Let α be an irrational in $(0,1)$ and let $\tau = R_\alpha$. Let $A_1 = [0,b)$ and $A_2 = [b,1)$ with $b \in (0,1)$. Let $(T_n(.))_{n\geqslant 1}$ be the random sequence of operators defined by the map

$$T(\omega) = \begin{cases} (M_{\phi_1})^* & \text{if} \quad \omega \in A_1 \\ (M_{\phi_2})^* & \text{if} \quad \omega \in A_2 \end{cases},$$

where $\phi_1, \phi_2 \in H^\infty(\mathbb{D})$ are nonconstant functions. Do Corollaries 3.22 and 3.23 hold for these sets A_1 and A_2, if $b \notin \mathbb{Z}\alpha$?

In our work, we only use two disjoint Borel subsets A_1 and A_2 of $[0,1)$ such that $A_1 \cup A_2 = [0,1)$ and $m(A_k) > 0$ for $k = 1, 2$. It is therefore natural to be interested in the case of more than two Borel subsets of $[0,1)$.

Open Problem 5.8. — Generalize our work on the random products of adjoints of multiplication operators for a finite number of disjoint Borel subsets A_k of $[0,1)$ covering $[0,1)$ such that $m(A_k) > 0$ for every $k \geqslant 1$, when the operator $T(\omega)$ is the adjoint of the multiplication operator $(M_{\phi_k})^*$ for every $\omega \in A_k$, with $\phi_k \in H^\infty(\mathbb{D})$ nonconstant.

Obviously, we can imagine similar results to Proposition 3.16 and Theorem 3.17 in the context of Problem 5.8. Finally, one can also look at the linear dynamics of random products $(T_n(.))_{n\geqslant 1}$ when the operator $T(\omega)$ is another type of operator on a space of holomorphic functions.

Open Problem 5.9. — Study the linear dynamics of random sequences $(T_n(.))_{n\geqslant 1}$ when the operator $T(\omega)$ is given by

$$T(\omega) = \begin{cases} T_1 & \text{if} \quad \omega \in A_1 \\ T_2 & \text{if} \quad \omega \in A_2 \end{cases},$$

where T_1, T_2 are two operators belonging to some specific families of operators on separable Fréchet spaces, where τ is either the doubling map or an irrational rotation on $\mathbb{T}$, and where A_1, A_2 are two disjoint Borel subsets of $[0,1)$ such that $A_1 \cup A_2 = [0,1)$ and $m(A_k) > 0$ for $k = 1, 2$.

For example, it would be natural to consider composition operators for T_1 and T_2 on a space of holomorphic functions and to study the linear dynamics of the corresponding sequence $(T_n(.))_{n\geqslant 1}$ in the situation of Problem 5.9, as well as weighted shifts on ℓ_p-spaces, since the linear dynamics of these operators is rather well known and of interest for many problems in linear dynamics.

The relevant centered function is

$$f = \mathbf{1}_{[0,b)} - \frac{b}{1-b}\mathbf{1}_{[b,1)} = \frac{1}{1-b}\big(\mathbf{1}_{[0,b)} - b\big). \tag{1}$$

Theorems 3.19 and 3.20 of the source imply both requested corollaries as soon as the Birkhoff sums of f have limsup $+\infty$ and liminf $-\infty$ almost everywhere.

2 Classical one-sided input

For an ergodic probability-preserving map T and a measurable set A, write

$$\Delta_n(A;x) = \sum_{j=0}^{n-1}\big(\mathbf{1}_A(T^j x) - \mu(A)\big).$$

Halász proved that if $\Delta_n(A;x)$ is bounded on one side in n for x in a set of positive measure, then $e^{2\pi i\mu(A)}$ is an eigenvalue of T. Sós's survey states this strengthening explicitly on printed page 234:

V.T. Sős: Irregularities of partitions 234

Concerning intervals of small discrepancy, first we remark that

$$\sup_N \Delta_N^+([0,\beta)) < \infty$$

may hold also in the case when $\beta \neq \{k\alpha\}$, i.e. when

$$\sup_N \Delta_N([0,\beta)) = \infty .$$

In Dupain & T. Sős (1978) the characterization of the intervals $[0,\beta)$ with

$$\sup_N \Delta_N^+([0,\beta)) < \infty$$

is investigated. Here we mention just one of the new phenomena: there exists an α for which the set $\{\beta \mid \sup_N \Delta_N^+([0,\beta)) < \infty\}$ is of power of continuum.

However, the assertion in Theorem remains true, if instead of boundedness of $\Delta_N(A)$ we suppose only one-sided boundedness. Halâsz (1976) proved that if

$$\sup_N \Delta_N^+(A;x) < \infty$$

holds on a set $X \subset \Omega$ of positive measure, then $e^{2\pi i\mu(A)}$ must be an eigenvalue of T.

7 *PARTITION-PROBLEMS FOR GENERAL HYPERGRAPHS*

While in the previous sections we had to make a strong selection because of the large variety and wide scope of the theorems, here we have only limited number of results. Though the problems for general hypergraphs arise from investigations of irregularities on different structures, they are interesting on their own too. To start with, we have to remark that there are almost no lower bounds on the discrepancy of partitions of hypergraphs.

The theorems below give upper bounds for the discrepancy of partitions on hypergraphs. One of the first results of this type is due to Olson & Spencer.

For completeness, here is a short coboundary proof of the precise bridge used in this packet.

Lemma 1 (one-sided bounded sums force a measurable coboundary)**.** *Let T be an ergodic probability-preserving transformation and let $g \in L^\infty$ be real with $\int g\,d\mu = 0$. If $\inf_{n\geq 0} S_n g(x) > -\infty$ almost everywhere, where $S_n g = \sum_{j=0}^{n-1} g \circ T^j$ and $S_0 g = 0$, then there is a finite measurable $G \geq 0$ such that*

$$g = G \circ T - G \quad \text{almost everywhere.}$$

The analogous assertion holds when the sums are bounded above.

Proof. Put

$$G(x) = \sup_{n\geq 0}\{-S_n g(x)\} < \infty.$$

Splitting the supremum into $n = 0$ and $n \geq 1$ gives the Bellman identity

$$G(x) = \max\{0, G(Tx) - g(x)\}.$$

Consequently

$$h := G + g - G \circ T \geq 0.$$

On $\{G > 0\}$ the two sides in the nonzero branch are equal, so $h = 0$. On $\{G = 0\}$ we have $h = g - G \circ T$, hence $0 \leq h \leq g^+$. Thus $h \in L^1$ and

$$g = G \circ T - G + h. \tag{2}$$

Summing (2) and dividing by n yields

$$\frac{G(T^n x)}{n} = \frac{S_n g(x)}{n} + \frac{G(x)}{n} - \frac{S_n h(x)}{n} \longrightarrow -\int h\,d\mu$$

almost everywhere, by ergodicity. The left side is nonnegative, while $h \geq 0$; therefore $\int h = 0$ and $h = 0$ almost everywhere. Equation (2) is the desired coboundary identity. Apply the same argument to $-g$ for upper boundedness. $\square$

3 Two-sided divergence for every nonexceptional interval

Theorem 1. *Let $\alpha \in (0,1) \setminus \mathbb{Q}$, let $R_\alpha x = x + \alpha$ on $\mathbb{T}$, and let $b \in (0,1)$ satisfy $b \notin \mathbb{Z}\alpha$ modulo one. Set*

$$d_b = \mathbf{1}_{[0,b)} - b.$$

Then for Lebesgue-almost every $x \in \mathbb{T}$,

$$\limsup_{n\to\infty} S_n d_b(x) = +\infty, \qquad \liminf_{n\to\infty} S_n d_b(x) = -\infty.$$

Proof. The set of points at which $(S_n d_b)$ is bounded below is invariant under R_α: shifting the starting point changes the sums by deleting the first term and adding a fixed finite quantity. By ergodicity, this set has measure zero or one. If it had measure one, the lemma would give $d_b = G \circ R_\alpha - G$ for a measurable real G. Exponentiation gives

$$e^{2\pi i d_b} = e^{-2\pi i b} = \frac{e^{2\pi i G \circ R_\alpha}}{e^{2\pi i G}}.$$

Thus $e^{-2\pi i b}$ would be an eigenvalue of R_α. The eigenvalues of an irrational rotation are exactly $e^{2\pi i k\alpha}$, $k \in \mathbb{Z}$, which would force $b \in \mathbb{Z}\alpha$ modulo one, a contradiction. Hence the sums are unbounded below almost everywhere. Applying the same argument to $-d_b$ shows that they are unbounded above almost everywhere. For a real sequence, these two statements are exactly the asserted infinite liminf and limsup. $\square$

The same conclusion follows directly from Halász's theorem: one-sided boundedness on the positive-measure invariant set would make $e^{2\pi i b}$ an eigenvalue, again contradicting $b \notin \mathbb{Z}\alpha$.

4 Answer to Question 5.7

Theorem 2 (strong form of the requested extension)**.** *Fix any irrational α and any $b \in (0,1)$ with $b \notin \mathbb{Z}\alpha$ modulo one. Let $A_1 = [0,b)$, $A_2 = [b,1)$ and define the random products from nonconstant $\phi_1, \phi_2 \in H^\infty(\mathbb{D})$ exactly as in Question 5.7. If either analytic alternative in Corollaries 3.22 and 3.23 of arXiv:2507.00186v3 holds, then the random sequence is topologically weakly mixing. No continued-fraction condition on α is needed.*

Proof. By (1), the source function is the positive scalar multiple $f = d_b/(1-b)$. Theorem 1 gives its required two-sided divergence almost everywhere. The source paper's Theorem 3.19 applies under the image-meets-the-unit-circle alternative, and Theorem 3.20 applies under the nonouter alternative. Their conclusions are precisely topological weak mixing of the random sequence. This proves both Corollary 3.22 and Corollary 3.23 in the setting of Question 5.7 and removes the arithmetic hypotheses that those corollaries used only to obtain a central limit theorem. $\square$

5 Scope and novelty audit

The statement is almost-everywhere, exactly matching the source's definition of a random sequence having the dynamical property. It is not true that every starting point must have two-sided divergence. Ying and Zheng (arXiv:2210.08441) give sharp arithmetic criteria, for rational interval lengths, under which a specified orbit has one-sided bounded discrepancy. Those exceptional orbits form a null invariant set here.

The bounded search covered the exact source title and arXiv id, the phrases "one-sided boundedness", "interval discrepancy", "Birkhoff sums bounded below", and "measurable coboundary", and the sources Halász (1976), Sós's survey, Ying–Zheng (2022), and Conze–Marco (arXiv:1302.3075). No later paper was found that explicitly says it answers Question 5.7. However, the exact one-sided eigenvalue implication needed for the answer is the classical Halász theorem. Therefore the correct provenance is `literature_implied_answer`, not a new full solution.

References

1. V. Gillet, *Linear dynamics of random products of operators*, arXiv:2507.00186v3; J. Functional Analysis (2026), article 111416.
2. G. Halász, *Remarks on the remainder in Birkhoff's ergodic theorem*, Acta Math. Acad. Sci. Hungar. 28 (1976), 389–395, doi:10.1007/BF01896805.
3. V. T. Sós, *Irregularities of Partitions*, in *Ramsey Theory, Uniform Distribution and Irregularities of Partitions*, Algorithms and Combinatorics 22, Springer, 2001; see printed p. 234 for the one-sided Halász theorem.
4. J. Ying and Y. Zheng, *On the one-sided boundedness of the local discrepancy of $\{n\alpha\}$-sequences*, Acta Arith. 206 (2022), 97–114; arXiv:2210.08441.
5. J.-P. Conze and J. Marco, *Remarks on step cocycles over rotations, centralizers and coboundaries*, arXiv:1302.3075.